Yasrib Yasir Farook

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EDUCATION

Eastern Michigan University Ypsilanti, MI
B.S. in Computer Science – Applied | *GPA: 3.4/4.0* *Expected Apr. 2027*

Relevant Coursework: Artificial Intelligence, Data Structures & Algorithms, Software Engineering, Applied Cryptography, Data Analytics, Database Principles, Operating Systems, Discrete Structures

RESEARCH INTERESTS

Generative AI | Large Language Models (LLMs) | Human–AI Collaboration | Machine Learning | Deep Learning | Medical AI | Human-Computer Interaction (HCI) | Computer Vision

RESEARCH EXPERIENCE

Investigating the Effects of Generative AI Training on User Story Writing Jan. 2026 – Present
Research Lead | *Faculty Advisor: Dr. Siyuan Jiang* *Eastern Michigan University*

- Designed a controlled empirical study investigating whether structured GenAI training improves users' interactions with LLMs and the quality of software requirements they produce.
- Developed a training intervention around four AI-interaction skills: context provision, prompt specificity, AI-output evaluation, and iterative refinement.
- Designed study scenarios, experimental procedures, pre/post assessments, control and intervention conditions, evaluation methodology, and the human-subject IRB protocol.

AI for Echocardiography using EchoNet-Dynamic Jul. 2026 – Present
Undergraduate Research Assistant | *Faculty Advisor: Dr. Yunting Yin* *Eastern Michigan University*

- Conducting medical AI research using EchoNet-Dynamic, containing 10,030 echocardiography videos with cardiac function measurements and ventricular annotations.
- Developed a Python pipeline linking 10,030 AVI videos with clinical metadata and 425,010 volume-tracing records; validated corresponding tracing data for 10,024 videos.
- Developing PyTorch-based preprocessing and deep learning pipelines for left ventricular segmentation, cardiac motion representation, and ejection-fraction prediction.

Using Generative AI for User Story Writing Mar. 2025 – Apr. 2026
Undergraduate Research Assistant | *Faculty Advisor: Dr. Siyuan Jiang* *Eastern Michigan University*

- Conducted empirical research on how Generative AI affects software-requirement quality in undergraduate software engineering education.
- Analyzed 406 user stories from 48 students at two U.S. public universities using INVEST-based measures, comparing requirements produced with and without GenAI assistance.
- Identified improvements in Testability ($0.651 \rightarrow 0.858$) and Structure ($0.630 \rightarrow 0.876$), while finding limitations in requirement decomposition and scope judgment.
- Contributed to a peer-reviewed ACM SIGCSE Virtual 2026 publication as second author.

Disability and Accessibility in Computer Science Education Jan. 2025 – Present
Undergraduate Research Assistant | *Faculty Advisor: Dr. Siyuan Jiang* *Eastern Michigan University*

- Conducting a systematic literature review on disability, accessibility, and inclusion in computer science education.
- Screened approximately 400 abstracts and 100 full-text papers using predefined inclusion/exclusion criteria and categorized empirical studies by methodology, research focus, and accessibility themes.

PUBLICATION

Brockenbrough, A., **Farook, Y. Y.**, Amore, T., & Jiang, S. (2026). "Guiding Students in Using Generative AI for User Story Writing: An Experience Report." *Proceedings of the 2nd ACM Virtual Global Computing Education Conference (SIGCSE Virtual 2026)*, ACM, 7 pages. **DOI:** 10.1145/3795867.3831003

CONFERENCE & RESEARCH PRESENTATIONS

Machine Learning Conference, EMU – Oral Presentation 2026
Can AI Understand the Beating Heart?
University Of Michigan AI Symposium – Poster Presentation 2026
Can AI Understand the Beating Heart?
MID-SURE 2026, Michigan State University – Poster Presentation 2026
Investigating the Effects of Generative AI Training on User Story Writing
Undergraduate Symposium, Eastern Michigan University – Poster Presentation 2026
Guiding Students in Using Generative AI for User Story Writing: An Experience Report

TECHNICAL & RESEARCH SKILLS

Programming: Python, Java, JavaScript, SQL, HTML/CSS, C, C++

AI/ML: PyTorch, Deep Learning, LLM Evaluation, Medical Computer Vision, AI-Assisted Software Engineering

Research: Experimental Design, Empirical Software Engineering, Human-Subject Studies, Systematic Literature Reviews, Statistical Analysis, IRB Protocol Development

Tools/Frameworks: Git, GitHub, React.js, Node.js, Express.js, MongoDB, VS Code

TEACHING EXPERIENCE

Computer Science Instructor, Engage EMU May – Aug. 2025

- Taught 100+ middle-school students HTML, CSS, JavaScript, and Python through hands-on, project-based computing activities.
- Guided students in programming, debugging, computational thinking, and problem-solving while adapting instruction to different experience levels.

Teaching Assistant, Neurodivergent Coding Camp May – Aug. 2026

- Mentored neurodivergent students in Python, block-based programming, and LEGO SPIKE Prime robotics through individualized, accessible instruction.

SCHOLARSHIPS & AWARDS

Open Science Award, University Of Michigan 2026
Departmental Recognition for Undergraduate Research Presentation 2026
2nd Place, Track Challenge, SpartaHacks 26 2026
Dean's List, Eastern Michigan University 2024, 2026
Diversity and Inclusion Leadership Award, Student Government 2024
Residential Scholar Award, Eastern Michigan University 2023
4WARD Scholarship 2023

LEADERSHIP

Speaker of the House, Eastern Michigan University Student Government 2025 – 2026
Led student legislative proceedings and represented student interests in university governance.
Student Senator, Eastern Michigan University Student Government 2023 – 2025
Represented students and supported initiatives related to student success, campus improvement, diversity, and engagement.
Founder & Secretary, Bangladesh Student Association 2025 – 2026
Founded the organization to strengthen cultural engagement and community building for Bangladeshi students.

ADDITIONAL TECHNICAL EXPERIENCE

IT Support Analyst – Desktop & Classroom Support, Eastern Michigan University Aug. 2025 – Present

Diagnose hardware, software, networking, audiovisual, and classroom-technology issues across university instructional environments.